Phase 1 — Data Acquisition

C1 Resource	C2 Equity	C3 Demand	C4 Infra	C5 Physical	ERP (C6 cand.) X
Global Solar Atlas v2 LTAy PVOUT, 1 km	NESDC/NSO poverty incidence	WorldPop 2020 100m	OSM power lines + cables	Hansen GFC + GHSL + DEM	Emission Reduction Potential (NTL proxy) **overlaps C3** **[EXCLUDED]**

Phase 2 — Pre-processing

Preprocessing & Normalisation

90 m resampling | C1, C3: pixel-level 2nd–98th percentile clip | C2: raw poverty % | Province-mean aggregation

Province level: C3 log(1 + 100x), then each criterion clipped at 2nd–98th percentile and scaled to [0, 1] once | ERP excluded (overlap with C3)

Phase 3 — EWM Weights

Entropy Weight Method (EWM)

Entropy: $e_j = -(\ln m)^{-1} \sum_i p_{ij} \ln p_{ij}$ Divergence: $d_j = 1 - e_j$ Weight: $w_j = d_j / \sum_k d_k$

Phase 4 — TOPSIS Scoring

TOPSIS: Priority Area Index (PAI)

Bounded ideal (unweighted) $A^+ = [1,1,1,1,1]$ $A^- = [0,0,0,0,0]$ | weighted: $v_j^+ = w_j$, $v_j^- = 0$

$$d_i^+ = \sqrt{\sum_j (v_{ij} - v_j^+)^2} \qquad d_i^- = \sqrt{\sum_j (v_{ij} - v_j^-)^2}$$

$$\mathrm{PAI}_i = d_i^- / (d_i^+ + d_i^-) \in [0,1]$$

Phase 5 — Spatial Analysis

Spatial Zoning (K-means k=5)

K-means k = 5 | n_init = 10 | Random seed fixed

Five priority zones ranked by mean PAI | cluster stability across k

LISA Autocorrelation

Queen contiguity weight matrix

Global Moran's I | Local Moran's I (LISA)

999 permutations | HH LL HL LH ns clusters

Phase 6 — Comparison

Deployment Comparison & Robustness

Deployment comparison against documented OSM solar installations

Zone-enrichment test (chi-square) | Weight sensitivity: equal and expert scenarios

Bootstrap weight uncertainty | Spatially restricted permutation tests

Phase 7 — Policy Screen

Counterfactual Allocation Rules + Zone-to-Policy Screen

S1	S2	S3	Screen
Population-proportional baseline	Equity-targeted (top 15 by PAI)	Resource-and-infrastructure comparator	Instruments examined, utility connection gates